13. Higher-order topology with explicit boundary and symmetry assumptions

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A topological material can have interesting boundary behavior even when its broad edges do not conduct. In certain models, charge accumulates in a quantized way near corners instead. This problem asks how to establish that behavior carefully: what property belongs to the bulk, what depends on the way the sample is cut, and what extra symmetry is needed to place a corner state at exactly zero energy?

Key terms

- **Higher-order topology** — Topology associated with lower-dimensional boundaries, such as corners of a two-dimensional sample or hinges of a three-dimensional sample.
- **Quadrupole** — A pattern of electric charge organization beyond total charge and ordinary dipole polarization; the selected model supports a quantized version under particular assumptions.
- **Bulk** — The interior of a system, far from boundaries.
- **Wilson loop** — A matrix describing how a chosen set of quantum states changes when transported around a closed path in momentum space.
- **Wannier sector** — A selected group of Wilson-loop eigenstates, separated from the others by a gap in that loop's spectrum.
- **Nested polarization** — A polarization calculated within such a Wannier sector, using a second transport calculation.
- **Corner excess charge** — Charge near a corner measured relative to a specified background, after accounting for other contributions.
- **Chiral symmetry** — An additional spectral symmetry that can relate positive and negative energies; it can matter for pinning special states to zero.
- **Termination** — The actual boundary cut and local boundary structure of the sample.

Why this matters

This problem broadens our understanding of how a material's interior organizes behavior at its boundaries. It also corrects a common interpretive shortcut: a quantized corner charge and a zero-energy corner state are not the same observation. Reliable distinctions are needed before such features can be used as design principles.

What progress would mean

Progress would produce a model with verified gaps, a well-defined bulk invariant and a carefully measured boundary consequence. It would establish a controlled example, rather than guarantee protected corner states for every sample shape or boundary.

Core question

For a fixed quadrupole-insulator family, can one certify the bulk and Wannier gaps, a quantized nested polarization, and the corresponding boundary charge under a specified termination?

Scope and assumptions

Start with a four-band Benalcazar–Bernevig–Hughes-type square-lattice model at half filling. Supply all hopping signs, orbital positions, mirror representations, and a rectangular termination. State whether chiral symmetry is imposed. Restrict parameters to a compact region away from bulk and Wannier-gap closings.

Separate fractional corner charge from an exactly zero-energy corner eigenstate. The latter requires additional spectral symmetry and boundary assumptions.

Mathematical target

Let P(k) be the rank-two occupied projector. Build the path-ordered Wilson loop W_x(k_y) from occupied-state overlaps. If its spectrum has a certified separation into Wannier sectors, form a smooth sector projector and its Berry holonomy along y. Define the nested polarization from that holonomy modulo one.

Do not replace a nested Wilson loop by the integral of the first Wilson-loop eigenphases. Define corner excess charge relative to an explicit ionic/reference background and a stated spatial window, with finite-size and edge contributions controlled.

Required outputs and proof obligations

Return H, symmetry operators and identities, bulk and Wannier-gap enclosures, nested invariant, boundary charge definition, and finite-size bounds. Any robustness statement must preserve the protecting symmetry and relevant gaps. State whether conclusions are intrinsic bulk statements or termination-dependent observables.

Validation and rejection controls

Recover topological and trivial limits of the selected model. Verify zero total Chern number where required while the nested invariant is nontrivial. Break a protecting symmetry as a control; the checker must stop claiming quantization. Compare two terminations and explain any boundary difference.

Milestones and research extension

Baseline: reproduce one quadrupole benchmark with valid Wilson-loop construction. Strong: certify its invariant and boundary charge on a parameter box. Research extension: classify a bounded family or one additional symmetry setting; no universal HOTI classification is implied.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Benalcazar, Bernevig and Hughes, Quantized Electric Multipole Insulators — Nested Wilson loops and assumptions for quantized multipole response.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$W_x(k_y) = \mathcal{P}\exp\left(i\oint \mathcal{A}_x(k_x,k_y)\,dk_x\right), \qquad \mathcal{A}_{x,ab} = i\langle u_a|\partial_{k_x}u_b\rangle.$$

$$\widetilde{W}_y = \mathcal{P}\exp\left(i\oint \widetilde{\mathcal{A}}_y\,dk_y\right), \qquad p_y^{\text{sector}} = \frac{1}{2\pi}\arg\det\widetilde{W}_y \pmod 1.$$

Use consistent holonomy orientation in an overlap implementation. The second connection belongs to a smoothly transported, spectrally isolated Wannier sector, not to the first loop's eigenphases alone. Prove both bulk and Wannier gaps and the relevant quantizing symmetry. Define corner charge with a reference background and termination; zero-energy corner states require additional assumptions.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — wrong observable (73–89).** The proposed quadrupole formula is a Chern-type curvature integral, not a quadrupole moment, and the corner-charge product formula is not universal. Replacing these prevents certifying the wrong invariant.

2. **Blocking — nested loop construction (78–85).** One must construct a Wannier-sector bundle and its holonomy; integrating eigenphases from the first loop is not that construction.
3. **Major — corner charge versus zero mode (101 onward).** Quantized boundary charge does not automatically pin a level to zero energy. Termination, filling, reference charge, and extra spectral symmetry matter.
4. **Major — scope (36–45).** All wallpaper/space-group HOTIs encompass different equivalence notions and boundary phenomena. The rewrite uses one well-specified benchmark and demands a Wannier gap before invoking nested polarization.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a fixed quadrupole-insulator family, can one certify the bulk and Wannier gaps, a quantized nested polarization, and the corresponding boundary charge under a specified termination?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: reproduce one quadrupole benchmark with valid Wilson-loop construction. Strong: certify its invariant and boundary charge on a parameter box. Research extension: classify a bounded family or one additional symmetry setting; no universal HOTI classification is implied.

Sources supporting the corrections

- Benalcazar, Bernevig and Hughes, Quantized Electric Multipole Insulators — Nested Wilson loops and assumptions for quantized multipole response.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What “pure thought” means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary

for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.